

SENTIMENT ANALYSIS FOR ENHANCED WELL-BEING USING NLP AND CHI-SQUARE MODELLING

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INTRODUCTION

In today's fast-paced digital age, understanding human emotions and sentiments expressed through textual data has become crucial across various domains ranging from social media analysis to customer feedback interpretation. This research project delves into the realm of sentiment analysis, employing advanced techniques and methodologies to unveil the emotional nuances embedded within textual and speech based responses and to find the relation between sentiment and sleep habits, time management and exercise using a chi-square model.

SDG 3: GOOD HEALTH AND WELL-BEING

The fusion of advanced sentiment analysis methodologies and statistical frameworks not only uncovers the intricate emotional tapestry within textual datasets but also holds profound implications for Sustainable Development Goal (SDG) 3: Good Health and Well-being. This innovative approach not only sheds light on emotional dynamics but also serves as a catalyst for prioritizing mental health and emotional well-being.



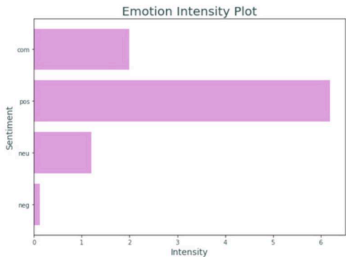
METHODOLOGY

- Leveraging state-of-the-art NLP models such as the VADER sentiment analyzer, this project aims to provide a comprehensive understanding of the sentiments conveyed in user-generated responses.
- Using the webkitSpeechRecognition() module to take speech based responses.
- Utilizing the pre trained NLP package, SentimentIntensityAnalyzer() to extract word polarity scores.
- Applying the Chi-Square model to determine the dependence between sentiments and time management, sleeping habits and physical activity.
- Exploring innovative visualization techniques, such as bar charts, to depict the distribution of emotions and sentiment scores effectively.

NLP (NATURAL LANGUAGE PROCESSING)

Natural Language Processing (NLP) is a field of artificial intelligence that focuses on enabling computers to understand, interpret and generate human language.

A means to use NLP to conduct sentiment analysis is through NLTK Vader in Python, which is a lexicon and rule based sentiment analysis tool specifically calibrated to sentiments most commonly expressed on social media platforms. When calculating a polarity score, Vader outputs four metrics: compound, negative, neutral and positive. The compound score calculates the sum of all lexicon ratings which is normalized between -1 (most negative) and +1 (most positive). Positive, negative, and neutral represent the proportion of the text that fall into these categories.



CONCLUSION

In conclusion, conducting sentiment analysis with NLTK and chi-square modelling in Python, we've gained valuable insights into the sentiments and mental health states of individuals. By collaborating with professional psychologists, personalized recommendations for activities have been tailored, fostering well-being in alignment with Sustainable Development Goal 3: Good Health and Well-being. This holistic approach not only enhances understanding of emotional well-being but also empowers individuals to prioritize their mental health, contributing to a healthier and more resilient society.

CHI-SQUARE MODELLING

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

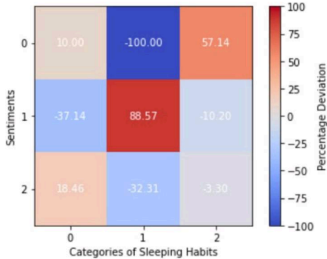
The chi-square test of independence is a statistical method used to determine whether there is a significant association between two categorical variables by comparing observed frequencies in a contingency table to those that would be expected if the variables were independent.

The `scipy.stats.chi2_contingency()` function in SciPy is utilized for conducting the chi-square test of independence on contingency tables, which assesses whether there exists an association between sentiments and the following variables -

- time management
- sleeping habits
- physical activity.

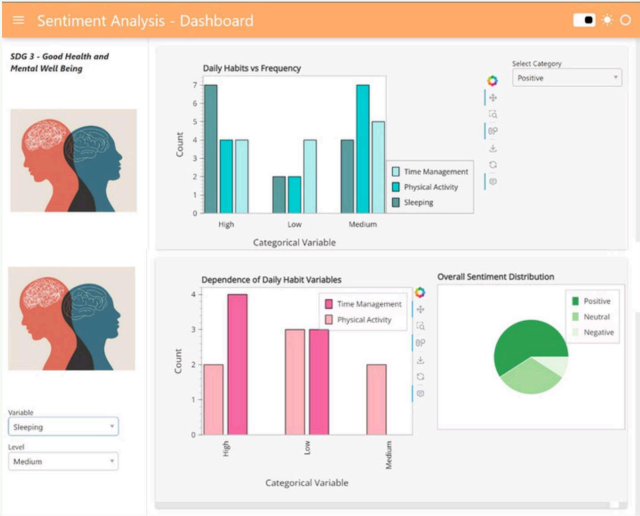
It takes the observed frequencies from the contingency table as input and computes the chi-square statistic along with its associated p-value. By comparing the computed p-value to a predetermined significance level, typically 0.05, we can either reject or accept our hypothesis.

Percentage Deviation Matrix for Sentiments and Sleeping Habits



RECOMMENDATIONS

After thorough analysis of our collected data and research through online resources, as well as consultation with a psychology professional, we listed a few recommendations for activities for the subjects of the study based on the scores that are produced from the analysis that could essentially help them in taking care of their mental health.



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